



國立臺灣大學
National Taiwan University

Lab02: CS50 Introduction

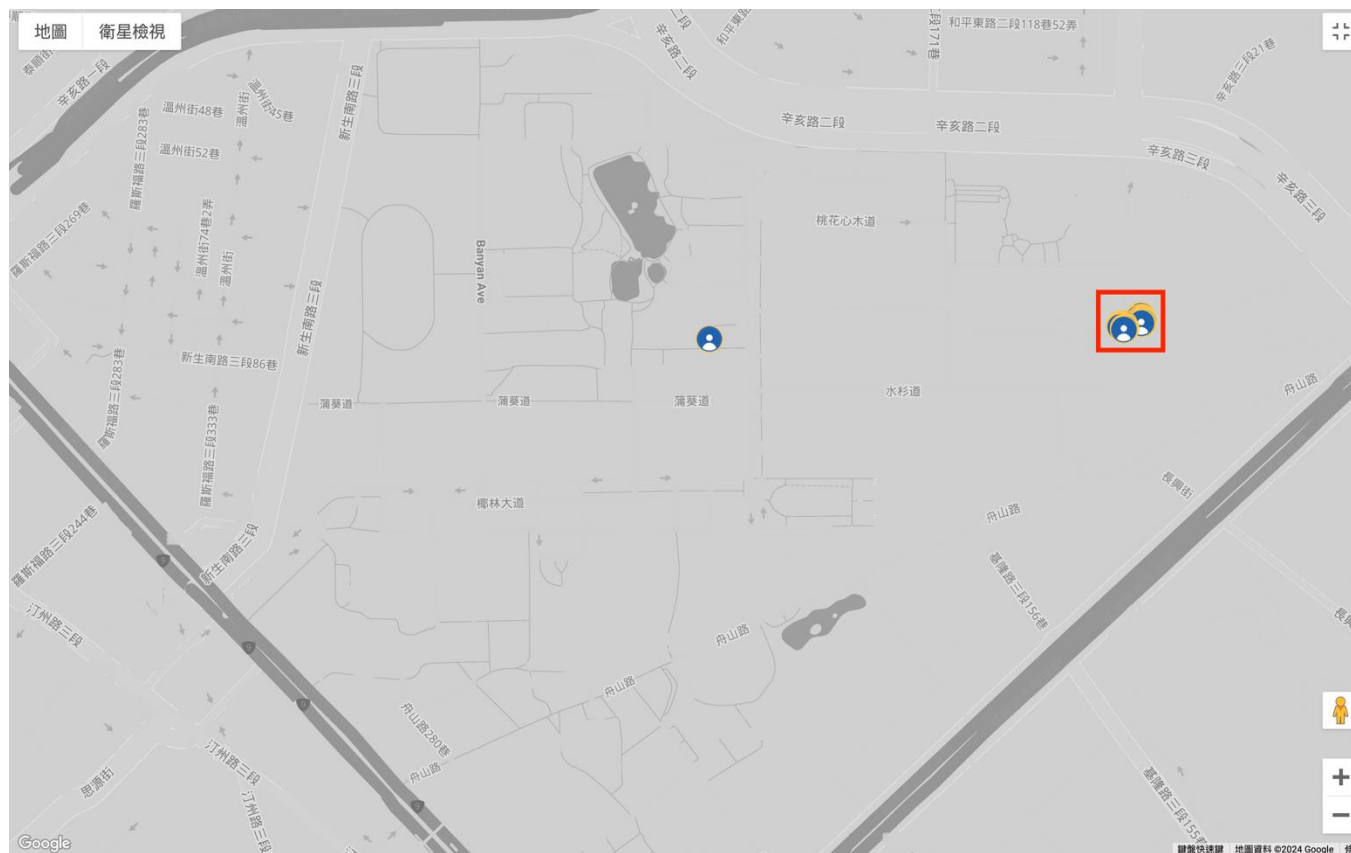
國立台灣大學 | 電機工程學系 | 林宗男 教授

重要通知

- 實驗課程 **LAB** 完成後，如果要**提早離開**，請同學必須**等待助教確認**後才可離開。
- **所有同學皆需把檔案上傳 NTU Cool**。請記得一定要繳交，否則將視同未完成實驗。
- **尚未加簽**「正課 + 實驗課」的同學完成實驗後務必找助教加簽。
- **離開教室時，請務必關機且螢幕不要關，最後也記得將椅子收齊。**

重要通知

- 請大家準時抵達電腦教室，若你的定位在紅匡之外，視同缺席。



重要通知

➤ 實驗課的出席狀況已公佈於 NTU Cool，大家每週都可以上去檢核自記得出席狀況。

2024_programming_實驗課出席狀況

檔案 編輯 查看 插入 格式 資料 工具 擴充功能 說明

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A1 學號

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3	B12202039	1											
4	B12204038	1											
5	B12401003	1											
6	B13901001	1											
7	B13901009	1											
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標記		評分規則
1	正常	
-2	未完成實驗	-7
-3	缺席	-10

CS50.ai

■ 1. CS50.ai 介紹

- CS50.ai 是哈佛大學計概課程推出的基於 ChatGPT 的程式設計助教，開放給所有人使用
- 它的名字叫做 duck debugger (ddb)，源自於 rubber duck debugging

CS50.ai 

CS50's adaptation of **ChatGPT** for students and teachers with support from **Microsoft** and **OpenAI**

beta

 Log In



CS50.ai

■ 1. CS50 介紹

- 使用前須以GitHub帳號登入
- 如果沒有GitHub帳號，則需要先辦一個



Sign in to **GitHub**
to continue to **CS50.ai**

Username or email address

Password

[Forgot password?](#)

Sign in

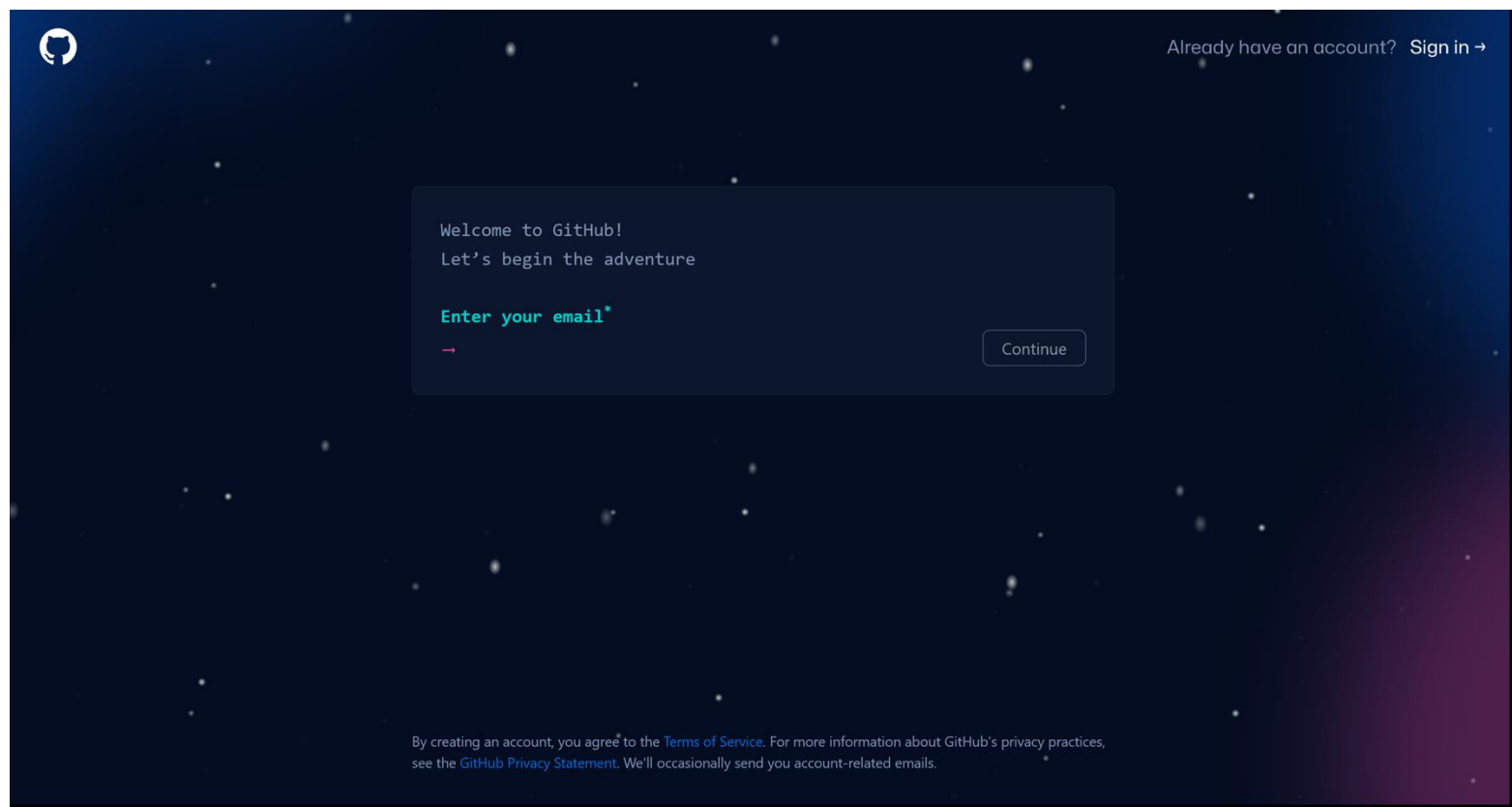
[Sign in with a passkey](#)

New to GitHub? [Create an account](#)

CS50.ai

■ 1. CS50 介紹

➤ GitHub帳號可以在官網辦



CS50.ai

■ 1. CS50 介紹

- 登入後的介面與 ChatGPT 相似
- 可以用中文或英文問問題，但它通常只會用英文回答

CS50.ai  

 Feedback

ddb

Quack. I am CS50's duck debugger (ddb), an experimental AI for [rubberducking](#). Quack quack. My replies might not always be accurate, so always think critically and let me know if you think that I've erred. Conversations are logged. Quack quack quack.

you

hello, how are you

ddb

Hello! How can I assist you with your CS50 or computer science questions today?

you

i want to know about print function in python

ddb

The **print** function in Python is used to output text or other data to the console. For example:

```
print("Hello, world!")
```

This will display "Hello, world!" on the screen. You can also print multiple items by separating them with commas:

```
print("Hello,", "world!")
```

What specific aspect of the **print** function would you like to know more about?

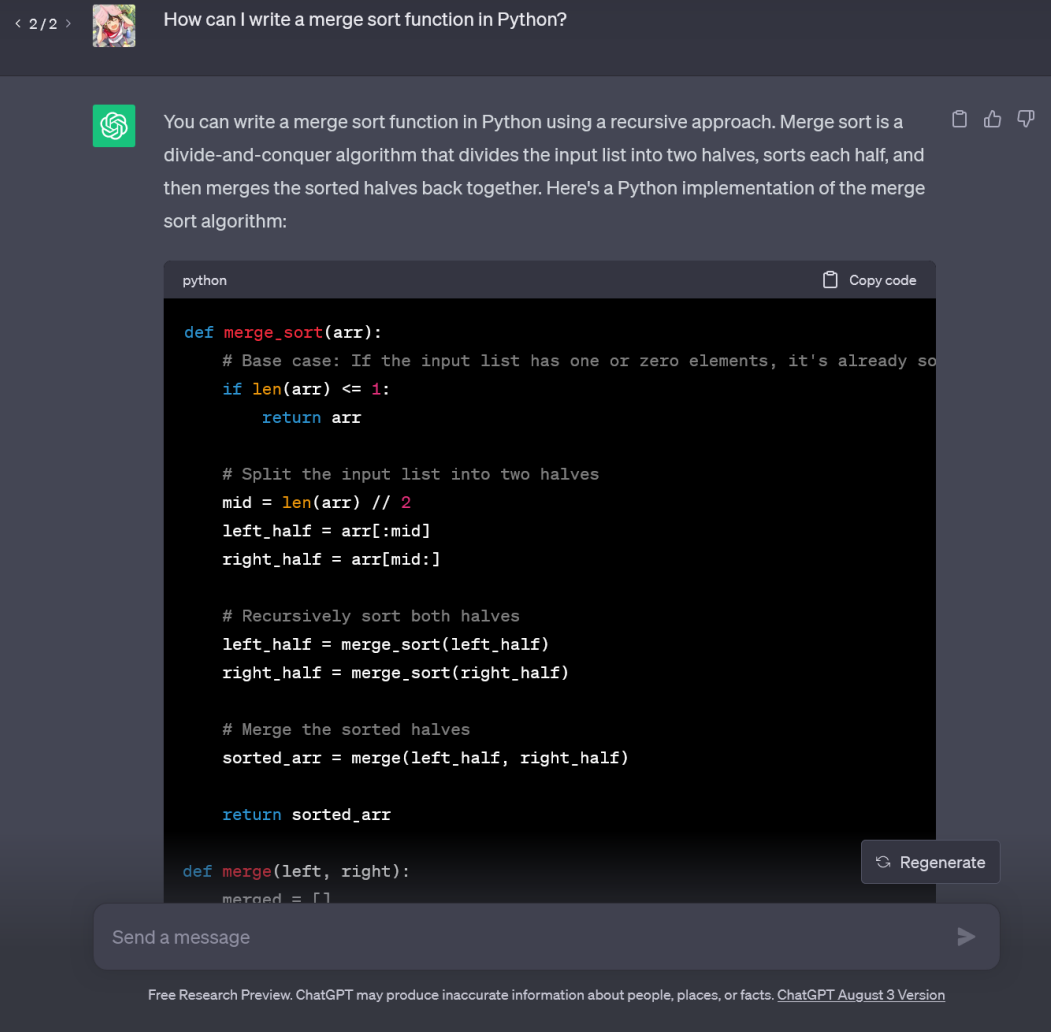
Ask a question



CS50.ai

■ 2. 和其他AI助理的差別

- 要求ChatGPT寫一段程式時，他會直接給出完整且可以直接使用的程式
- 在GitHub Copilot寫一段註解，他也會根據註解寫出完整的程式



The screenshot shows a ChatGPT chat window. The user's prompt is "How can I write a merge sort function in Python?". The AI's response explains the divide-and-conquer algorithm and provides a Python implementation. The code is displayed in a dark-themed editor with a "Copy code" button. At the bottom, there is a "Send a message" input field and a "Regenerate" button. A footer note states: "Free Research Preview. ChatGPT may produce inaccurate information about people, places, or facts. ChatGPT August 3 Version".

```
python
def merge_sort(arr):
    # Base case: If the input list has one or zero elements, it's already sorted
    if len(arr) <= 1:
        return arr

    # Split the input list into two halves
    mid = len(arr) // 2
    left_half = arr[:mid]
    right_half = arr[mid:]

    # Recursively sort both halves
    left_half = merge_sort(left_half)
    right_half = merge_sort(right_half)

    # Merge the sorted halves
    sorted_arr = merge(left_half, right_half)

    return sorted_arr

def merge(left, right):
    merged = []
```

CS50.ai

■ 2. 和其他AI助理的差別

- 而要求CS50.ai做相同的動作時，他會挖掉某些部分留給你自己完成，或是不給任何程式碼，而是**列出步驟引導你完成**
- 這麼做是為了避免直接給出答案，使學生失去思考能力

```
you
How can I write a merge sort function in Python?

ddb
Merge sort is a divide-and-conquer algorithm that works by splitting a list into two halves, sorting them, and then merging them back together. Here's a general idea of how you can implement it in Python:

1. If the list is of length 0 or 1, it is already sorted. So, return the list.
2. Otherwise, split the list into two halves.
3. Recursively sort both halves.
4. Merge the two sorted halves back together.

In terms of code, you might start with something like this:

def merge_sort(list):
    # Base case: single or empty list
    if len(list) <= 1:
        return list

    # Recursive case: split list into two halves
    mid = len(list) // 2
    left_half = merge_sort(list[:mid])
    right_half = merge_sort(list[mid:])

    # Merge the sorted halves
    # You'll need to write a helper function for this

Remember, you'll need to write a helper function to merge the sorted halves. This function should take in two sorted lists and return a single sorted list that contains all the elements from both input lists.

Ask a question
```